MODULE SERVER - CLOUDSTUDIO PLATFORM

Technical Specification & Competition Assessment Guidelines

INTRODUCTION

CloudStudio is a **high-performance infrastructure management and DevOps showcase platform**. The system allows software developers, system architects, and DevOps engineers to apply for developer identity and income validation, build **dynamic microservice infrastructure documentation** pages using predefined section templates, and apply for bare-metal server cluster financing or GPU cluster lease plans.

DESCRIPTION OF PROJECTS AND TASKS

Develop a robust RESTful API using Laravel 11.x and Laravel **Sanctum** for API token management. All responses returned by the application must strictly adhere to the application/json content type and emit semantic HTTP status codes.

Technical Baseline:

- Framework: Laravel 11.x
- Authentication: Laravel Sanctum
- Storage Architecture: Relational Database (MySQL)
- Target Subdirectory: `XX_SERVER_MODULE/BACKEND`

Phase One – CloudStudio REST API

A1: Authentication

The authentication architecture implements a multi-guard system to clearly differentiate administrative auditors from registered platform developers.

A1a: Register

POST `/api/v1/auth/register`

Request Headers

Header	Value
Accept	`application/json`
Content-Type	`application/json`

Body (JSON)

Field	Description / Rules
`full_name`	required, string
`username`	required, string, unique
`password`	required, string, min:6

A1a: Response Success

Status Code	Body

201 Created	<pre>{
	  "message": "Registration successful",
	  "user": {
	  "id": 1,
	  "full_name": "Alex Mercer",
	  "username": "alexm",
	  "role": "developer"
	  },
	  "token": "1|qgV2cTisoRVSwX6R2iPpxd..."
	}</pre>

A1b: Response Validation Failure

Status Code	Body

422 Unprocessable Content	<pre>{
	  "message": "Invalid field",
	  "errors": {
	  "username": [
	  "The username has already been taken."
	  ]
	  }
	}</pre>

A1c: Login

POST `/api/v1/auth/login`

Request Headers

Header	Value
Accept	`application/json`
Content-Type	`application/json`

Body (JSON)

Field	Description / Rules
`username`	required, string
`password`	required, string

A1c: Response Success

Status Code	Body

200 OK	<pre>{
	  "message": "Login successful",
	  "user": {
	  "id": 1,
	  "username": "admin_auditor",</pre>

Status Code	Body
	<pre> "role": "auditor" }, "token": "2|hZb7bB9cQx1K..." }</pre>

A1d: Response Incorrect Credentials

Status Code	Body
401 Unauthorized	<pre>{ "message": "Username or password incorrect" }</pre>

A1e: Logout

POST `/api/v1/auth/logout`

Request Headers

Header	Value
Accept	`application/json`
Authorization	`Bearer <TOKEN>`

A1e: Response Success

Status Code	Body
200 OK	<pre>{ "message": "Logout success" }</pre>

A1f: Response Invalid Token

Status Code	Body
401 Unauthorized	`{ "message": "Unauthorized user" }`

A2: Data Validation Gate

Developers must undergo compliance review to unlock core platform services such as publishing infrastructure pages and applying for server leasing.

A2a: Request Data Validation

POST `/api/v1/validation`

Request Headers

Header	Value
Accept	`application/json`
Content-Type	`application/json`
Authorization	`Bearer <TOKEN>`

Body (JSON)

Field	Description / Rules
`github_url`	required, url
`tech_stack`	required, string
`monthly_income`	required, numeric

A2a: Response Success

Status Code	Body
200 OK	`{ "message": "Request data validation sent successful" }`

A2b: Response Invalid Token

Status Code	Body
401 Unauthorized	<pre>{ "message": "Unauthorized user" }</pre>

A2c: Response Invalid Fields

Status Code	Body

422 Unprocessable Content	<pre>{
	  "message": "Invalid field",
	  "errors": {
	  "monthly_income": [
	  "The monthly income field must be a valid number."
	  ]
	  }
	}</pre>

A2d: Response Validation Already Submitted

Status Code	Body
400 Bad Request	<pre>{ "message": "A validation request has already been submitted and is either pending or already approved" }</pre>

A2e: Get Validation

GET `/api/v1/validations`

Request Headers

Header	Value
Accept	`application/json`

Header	Value
Authorization	`Bearer <TOKEN>`

A2e: Response Success

Status Code	Body

200 OK	{
	"validation": {
	"id": 1,
	"status": "pending",
	"github_url": "https://github.com/alexmercer",
	"tech_stack": "Kubernetes, Go, Rust, Terraform",
	"monthly_income": 4500.00,
	"auditor_notes": null,
	"auditor": null
	}
	}

A2f: Response Invalid Token

Status Code	Body

401 Unauthorized	{
	"message": "Unauthorized user"
	}

A2g: Auditor Review

PUT `/api/v1/validations/:id/review`

Request Headers

Header	Value
Accept	`application/json`
Content-Type	`application/json`

Header	Value
Authorization	`Bearer <AUDITOR_TOKEN>`

Body (JSON)

Field	Description / Rules
`status`	required, in:accepted,declined
`auditor_notes`	nullable, string

A2g: Response Success

Status Code	Body
200 OK	`{ "message": "Validation status updated successfully" }`

A2h: Response Forbidden

Status Code	Body
403 Forbidden	`{ "message": "Auditor access required" }`

A3: Infrastructure Page Builder

Allows developers to dynamically define their microservice and infrastructure architectural diagrams using schema-driven component sections.

A3a: Create Page

POST `/api/v1/pages`

Request Headers

Header	Value
Accept	`application/json`

Header	Value
Content-Type	`application/json`
Authorization	`Bearer <TOKEN>`

Body (JSON)

Field	Description / Rules
`title`	required, string
`slug`	required, string, unique

A3a: Response Success

Status Code	Body

201 Created	<pre>{
	  "message": "Page created successfully",
	  "page": {
	  "id": 1,
	  "title": "Edge Gateway Cluster",
	  "slug": "edge-gateway-cluster",
	  "is_published": false
	  }
	}</pre>

A3b: Response Validation Failure

Status Code	Body

422 Unprocessable Content	<pre>{
	  "message": "Invalid field",
	  "errors": {
	  "slug": [
	  "The slug has already been taken."
	  ]
	  }
	}</pre>

A3c: Append Section with Auto-Reordering

POST `/api/v1/pages/{slug}/sections`

Request Headers

Header	Value
Accept	`application/json`
Content-Type	`application/json`
Authorization	`Bearer <TOKEN>`

Body (JSON)

Field	Description / Rules
`template_id`	required, integer, exists:templates,id
`position`	required, integer, min:1

Technical Requirement: If the target position is already occupied by existing sections, the system must automatically increment the positions of conflicting sections (position = position + 1) in an atomic database transaction.

A3c: Response Success

Status Code	Body
201 Created	`{ "message": "Section appended successfully", "section": { "id": 10, "template_id": 3, "position": 1 } }`

A3d: Response Not Found

Status Code	Body
404 Not Found	`{ "message": "Page not found" }`

A3e: Update Section Fields

PUT `/api/v1/pages/{slug}/sections/:section_id/fields`

Request Headers

Header	Value
Accept	`application/json`
Content-Type	`application/json`
Authorization	`Bearer <TOKEN>`

Body (JSON)

Field	Description / Rules
`fields`	required, array
`fields.*.template_field_id`	required, integer
`fields.*.value`	required, string

A3e: Response Success

Status Code	Body
200 OK	`{ "message": "Fields persisted successfully" }`

A3f: Response Not Found

Status Code	Body

404 Not Found	{
	"message": "Section not found"
	}

A3g: Publish Infrastructure Page

PUT `/api/v1/pages/{slug}/publish`

Request Headers

Header	Value
Accept	`application/json`
Authorization	`Bearer <TOKEN>`

A3g: Response Success

Status Code	Body

200 OK	{
	"message": "Page published successfully"
	}

A3h: Response Profile Not Accepted

Status Code	Body

403 Forbidden	{
	"message": "Developer profile validation is not accepted yet. Cannot publish page."
	}

A3i: Public Infrastructure Page Discovery

GET `/api/v1/public/pages/{slug}`

Request Headers

Header	Value
Accept	`application/json`

A3i: Response Success

Status Code	Body

200 OK	{
	  "page": {
	  "title": "Edge Gateway Cluster",
	  "slug": "edge-gateway-cluster",
	  "developer": {
	  "full_name": "Alex Mercer",
	  "github_url": "https://github.com/alexmercer"
	  },
	  "sections": [
	  {
	  "id": 10,
	  "position": 1,
	  "template_id": 3,
	  "fields": [
	  {
	  "template_field_id": 12,
	  "value": "10.0.0.0/16 Virtual Private Cloud"
	  }
	  ]
	  }
	  ]
	  }
	}

A3j: Response Not Found or Unpublished

Status Code	Body
404 Not Found	<pre>{ "message": "Page not found or not published" }</pre>

A4: Server Cluster Financing & Lease

Facilitates subsidized and enterprise hardware leasing for high-density compute tasks.

A4a: Get Server Clusters with Filters & Sorting

GET `/api/v1/server_clusters`

Request Headers

Header	Value
Accept	`application/json`

Query Parameters

Parameter	Description / Rules
`category`	optional, in:bare_metal,gpu_cluster
`max_price`	optional, numeric
`sort_by`	optional, in:price,name (default: price)
`order`	optional, in:asc,desc (default: asc)

A4a: Response Success

Status Code	Body

200 OK	<pre>{
	  "clusters": [
	  {
	  "id": 4,
	  "name": "NVIDIA H100 8x SXM5 Cluster",
	  "category": "gpu_cluster",</pre>

Status Code	Body
	<pre>"price": "4800.00", "available_months": [6, 12, 24, 36] }] }</pre>

A4b: Response Invalid Query Parameters

Status Code	Body
422 Unprocessable Content	<pre>{ "message": "Invalid query parameters" }</pre>

A4c: Apply Financing

POST `/api/v1/applications`

Request Headers

Header	Value
Accept	`application/json`
Content-Type	`application/json`
Authorization	`Bearer <TOKEN>`

Body (JSON)

Field	Description / Rules
`cluster_id`	required, integer, exists:server_clusters,id
`months`	required, integer
`notes`	nullable, string

A4c: Response Success

Status Code	Body

201 Created	<pre>{
	  "message": "Applying for server cluster financing successful",
	  "application": {
	  "id": 1,
	  "cluster_id": 4,
	  "months": 12,
	  "monthly_nominal": "400.00",
	  "status": "pending"
	  }
	}</pre>

A4d: Response Invalid Token

Status Code	Body
401 Unauthorized	<pre>{ "message": "Unauthorized user" }</pre>

A4e: Response Validation Gate Not Accepted

Status Code	Body
403 Forbidden	<pre>{ "message": "Your profile validation must be accepted by auditor before applying" }</pre>

A4f: Response Invalid Fields

Status Code	Body

422 Unprocessable Content	<pre>{
	  "message": "Invalid field",
	  "errors": {
	  "cluster_id": [
	  "The selected cluster id is invalid."</pre>

Status Code	Body
	], "months": ["The selected months tenure is not available for this server cluster."] } }

A4g: Response Income Criteria Failed

Status Code	Body

422 Unprocessable Content	{
	"message": "Monthly income insufficient for this server cluster lease plan"
	}

A4h: Response Active Lease Exists

Status Code	Body

400 Bad Request	{
	"message": "User already holds an active lease agreement"
	}

A4i: Get Developer Lease Applications

GET `/api/v1/applications`

Request Headers

Header	Value
Accept	`application/json`
Authorization	`Bearer <TOKEN>`

A4i: Response Success

Status Code	Body

200 OK	{
	"applications": [
	{
	"id": 1,
	"cluster": {
	"id": 4,
	"name": "NVIDIA H100 8x SXM5 Cluster",
	"category": "gpu_cluster",
	"price": "4800.00"
	},
	"months": 12,
	"monthly_nominal": "400.00",
	"status": "pending",
	"notes": "Dedicated training node for local LLM inference.",
	"created_at": "2026-11-05 12:00:00"
	}
	]
	}

A4j: Response Invalid Token

Status Code	Body

401 Unauthorized	{
	"message": "Unauthorized user"
	}

Phase Two – Advanced Assessment

B1: Availability Flagging & Stock Reservation

The server cluster catalog must include stock availability flags (`stock` and `is_available`). When a developer applies for a lease, the system deducts the stock atomically. If the stock reaches 0, the system must reject the application.

B1a: Cluster Stock Depleted

POST `/api/v1/applications`

Request Headers

Header	Value
Accept	`application/json`
Content-Type	`application/json`
Authorization	`Bearer <TOKEN>`

Body (JSON)

Field	Description / Rules
`cluster_id`	required, integer, exists:server_clusters,id
`months`	required, integer
`notes`	nullable, string

B1a: Response Cluster Out of Stock

Status Code	Body
409 Conflict	`{ "message": "Selected server cluster is currently out of stock" }`

B2: Dynamic Template Validation

Section field values must strictly adhere to the dynamic field schema defined by each template (`ipv4_cidr`, `port`, `url`).

B2a: Dynamic Template Validation

PUT `/api/v1/pages/{slug}/sections/:section_id/fields`

Request Headers

Header	Value
Accept	`application/json`
Content-Type	`application/json`
Authorization	`Bearer <TOKEN>`

Body (JSON)

Field	Description / Rules
`fields`	required, array
`fields.*.template_field_id`	required, integer
`fields.*.value`	required, valid format per template schema

B2a: Response Dynamic Validation Format Fails

Status Code	Body
422 Unprocessable Content	`{ "message": "Validation error for dynamic field", "errors": { "fields.0.value": [ "The value must be a valid IPv4 CIDR notation (e.g. 10.0.0.0/16)." ] } }`

B3: Debt Service Ratio (DSR) Enforcement

A financial solvency assessment mechanism enforcing a maximum Debt Service Ratio of 40% of the verified monthly income (`monthly_income * 0.4`), calculated cumulatively with any other lease applications currently in `active` status.

B3a: Debt Service Ratio (DSR) Enforcement

POST `/api/v1/applications`

Request Headers

Header	Value
Accept	`application/json`
Content-Type	`application/json`
Authorization	`Bearer <TOKEN>`

Body (JSON)

Field	Description / Rules
`cluster_id`	required, integer, exists:server_clusters,id
`months`	required, integer
`notes`	nullable, string

B3a: Response DSR Exceeded

Status Code	Body
422 Unprocessable Content	`{ "message": "Total monthly lease commitment exceeds the 40% Debt Service Ratio (DSR) limit of your verified income" }`